

Analytics of Dining Activity Across American University Campus



Report for

ITEC-620

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1. Executive Summary

The project analyzes the dining activity of American University during the Spring 2025 semester, through various food venues by means of data transactions that were taken directly from Grubhub's OneView platform.

The analysis's aim is to unravel the ways in which students, employees, and other community members utilize the dining options provided on campus, to recognize the venues and menu items that are performing the best, and to find out the already existing trends that can make the choices of staffing, inventory control, and operations easier. We made use of descriptive, forecasting, and regression statistical methods to conduct the analysis.

Key findings include:

- **Top Venues:** Panera Bread, Qdoba, Starbucks, and Subway have the highest order volumes, with Panera Bread leading at 84,462 total items amongst a total of 284,597 items ordered across all venues.
- **Popular Menu Items:** Qdoba's "Bowl" was the single most ordered item (17,049 items), followed by Panera Bread's "Half Sandwich and Cup of Soup" (12,531 items).
- **Temporal Patterns:** April had the highest total orders (82,476), with weekdays significantly busier than weekends. January and May saw lower activity due to semester start and end.
- **Predictive Insights:** Number of items sold at Panera exhibited marginal dependence on weather and nutritional factors.

These findings provide actionable insights to optimize staffing, inventory, and menu planning across campus dining operations.

2. Introduction

The campus dining services must deal with the difficult task of providing services to a huge number of customers without sacrificing quality, and at the same time, managing their stocks in a way that will not

lead to wastage. Therefore, it is important to have a thorough understanding of how the different food venues are utilized by students, staff, and visitors to ensure the availability of food, the quality of service, and the reduction of operational waste. The project will deal with the analysis of the dining activity at American University for the period of the spring 2025 semester with the purpose of identifying patterns in the usage of different venues, the popularity of menu items, and the fluctuations in demand over time. The institution uses the Grubhub OneView platform to perform transaction management over various dining locations.

By analyzing this dataset, the project seeks to identify:

1. High-traffic venues and menu items that require prioritized staffing and inventory management.
2. Temporal trends across the days of the week and months to optimize operational planning.
3. Predictive insights on order volume across venues and items to forecast future demand.
4. External factors (e.g. weather, nutrition) that may influence demand negatively or positively.

The insights derived from this analysis aim to support data-driven operational decisions, enhance customer satisfaction, and improve efficiency across campus dining services.

3. Data

3.1 Product Mix Data

This dataset contains records of items purchased across campus dining venues over a five-month period (from 1/3/2025 to 5/30/2025). A few sample records from the dataset are presented in Appendix Figure 1. Each record includes Date, Venue, Item, Category, and Quantity.

Data Cleaning and Preparation:

1. **Formatting:** Transaction dates were standardized to allow accurate aggregation by day, week, and month.
2. **Duplicate values:** Duplicate names of Venues had to be combined for the purpose of this analysis. For example, venues named “Subway – Meal Exchange” were renamed as “Subway”.

3. **Aggregation:** Data was summarized at the venue, item, day, and month levels to enable both descriptive and predictive analysis.
4. **Validation:** Aggregated totals were cross-checked with raw data to ensure accuracy.
5. **Ignored Columns:** Price, Subtotal and Total columns are ignored as items purchased through meal exchange program are recorded as zero cost items. Hence, revenue-related analyses could not be conducted using this dataset. Modifier columns were ignored for this project. For example, types of cheese, dressing, or bread are modifiers to a Subway sandwich.

3.2 Nutrition Data

1. Nutrition data of items sold [at Panera Bread](#) (highest selling venue) were used for this analysis.
2. It contains Caloris, Fat (grams), Protein (grams), Carbs (grams), and Sodium (mgrams) of 176 items sold at Panera Bread.
3. Nutrition information about a few sample items is shown in Appendix Figure 2. Leading and lagging whitespaces present in the item labels had to be removed before data usage.

3.3 Weather Data

1. Daily weather data from 1/1/2025 to 5/30/2025 of Washington DC was obtained from [Visual Crossing](#).
2. A sample of the dataset is shown in Appendix Figure 3. It contains information about temperature, humidity, precipitation, snow, wind, etc. No modifications were made to the data before usage.

4. Analysis

The dataset summary is as follows

Total Venues	Total Categories	Total Unique Items	Total Item Quantity
10	78	624	284,597

The semester starts out slowly and tapers out towards the end. In the middle of the semester, February through April can be characterized as months with a steady stream of orders. On an average day, 1500 to 3000 items were purchased across all venues, which averages 15000 to 20000 in a week and 60000 to 80000 items in an average month.

The median of total items sold on a day is 2,407 with a mean of 2,047 items. The distribution of daily orders is right-skewed, meaning there are several low-volume days (Spring break and semester start/end) pulling the mean downward. Typical days are actually busier than what the mean suggests.

4.2 Venue-Level Analysis

Total item quantities (across all venues and all days) were aggregated by venue to determine utilization levels. Panera Bread was the most visited venue, reflecting strong popularity. Qdoba, Starbucks, and Subway also show significant traffic. These insights help prioritize staffing and inventory allocation. Panera Bread alone is considered for detailed analysis (e.g. forecasting and regression studies).

Top 10 Venues by Total quantity:

Venue	Total Quantity
Panera Bread	84,462
Qdoba	40,001
Starbucks	33,966
Subway	30,347
Freshens	28,969
True Burger	26,159
Terrace Dining Room	23,734
Baba's Pizza	11,000
ACE Sushi	5,957

A time series comparison of the top three venue's showcases Panera Bread's relative volatile nature of their quantity of items sold when in comparison to Qdoba and Starbucks(Figure 17), implying Panera Bread is affected more by the day of the week.

4.3 Item-Level Analysis

Total item quantities (across all venues and all days) were aggregated by venue and item to identify the most frequently purchased menu items. Qdoba and Panera Bread dominate top items, highlighting the importance of customizable entrées and combo meals. True Burger’s inclusion in the top 10 indicates competitive performance in key menu categories.

Top 10 Items Across All Venues:

Venue	Item	Total Quantity
Qdoba	Bowl	17,049
Panera Bread	Half Sandwich and Cup of Soup	12,531
True Burger	3 Piece Chicken Tenders	12,154
Qdoba	Burrito	10,912
Qdoba	Grilled Quesadilla	10,892
Panera Bread	Mac & Cheese and Half Sandwich	9,989
Baba’s Pizza	One Topping Pizza	7,623
Panera Bread	Half Sandwich and Half Salad	5,961
True Burger	Classic Americana Burger	4,859
Panera Bread	Large Mac & Cheese	4,203

Viewing the top three items’ quantities sold on a time-series graph (figure 18) further supports the Qdoba bowls’ dominance over the other items—not once did the half sandwich/soup, nor the 3-piece chicken tenders ever outcompete the Qdoba bowl, regardless of the date. We also see a long-term switch between the second and third most popular item—Panera's half sandwich & cup of soup was the second most popular item from January and February, but from March to May, the 3-piece chicken tenders deal realizes higher quantities sold. Additional analysis is required to determine the exact cause, if any.

4.4 Day-Level Analysis

Total item quantities (across all venues) were aggregated by date to identify the top 10 days. The numbers throughout the Top 10 days vary from 3,323 to 3,545, and the most popular day was March 31. Five of the top days belong to April, whereas March has three and all of the days were on weekdays.

Top 10 Highest Volume Days Across All Venues:

Date	Quantity
2025-03-31	3,545
2025-04-17	3,462
2025-03-27	3,420
2025-04-28	3,415
2025-04-14	3,412
2025-02-13	3,409
2025-04-21	3,392
2025-04-22	3,392
2025-04-24	3,390
2025-01-30	3,323

Overall, a knowledge of peak consumption days informs staffing and inventory needs for operational planning. Retrospectively, we would be interested in identifying if this peak activity is correlated to any on-campus events (e.g. sports, campus tours).

4.5 Day of the Week Analysis

Day of the week analysis was used to identify days with the highest traffic. Weekday orders are significantly higher than weekends, consistent with student schedules and campus activity. Additional analysis on how the day of the week impacts the total quantity of output by venue can be found in the forecasting section.

Days	Total Quantity	Mean Daily Quantity	Median Daily Quantity
Monday	49,801	2490	3202
Tuesday	49,108	2455	3122
Wednesday	46,752	2338	3026
Thursday	49,020	2334	3186
Friday	36,976	1761	2417
Saturday	23,543	1239	1605
Sunday	29,397	1633	1930

4.6 Month-Level Analysis

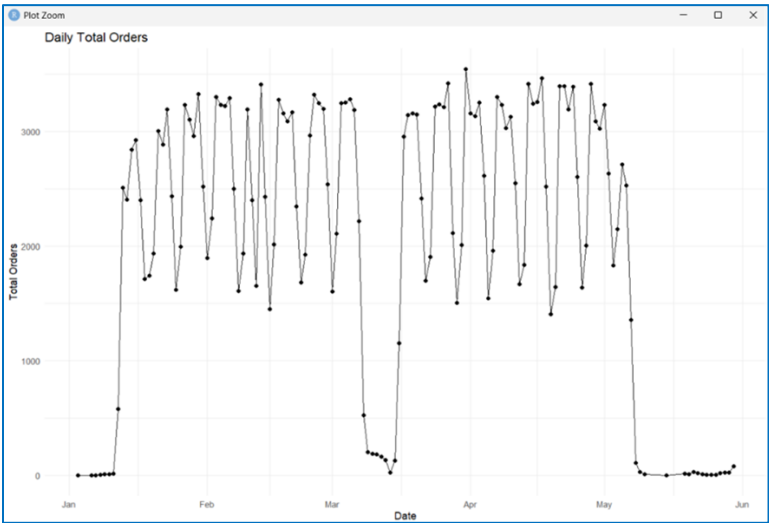
April is the peak month, followed by February and March. January and May show lower volume due to semester start/end and breaks. These patterns align with campus occupancy trends. March activity is low due to the spring break.

Month / Year	Total Quantity
April 2025	82,476
February 2025	73,672
March 2025	62,267
January 2025	49,343
May 2025	16,839

5. Time series

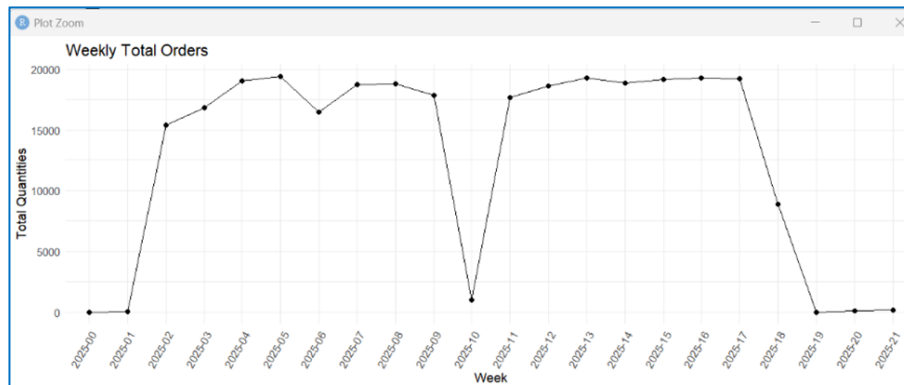
A time series analysis was performed on the dataset to reveal the temporal patterns in the dining activity throughout the semester. Long-term trends and short-term fluctuations were captured using the three time-based views daily, weekly, and monthly.

The daily time series reveals a very regular weekly seasonality pattern plus a distinct break period. The pattern confirms that campus dining is subject to regularities connected with class schedules, with demand peaking on Monday–Thursday and dropping during the weekends. This makes it very easy to forecast daily operations during the active weeks.



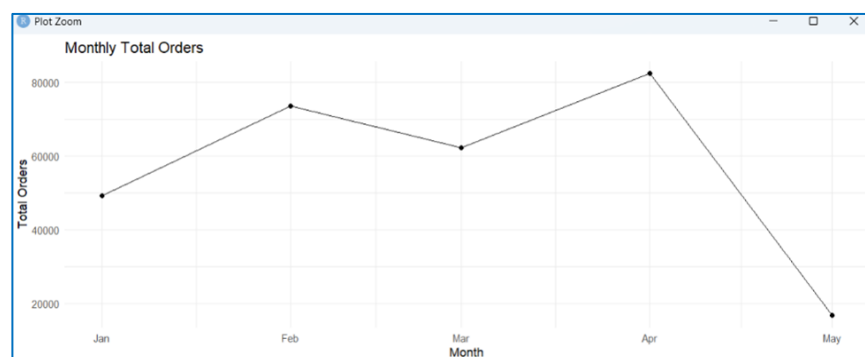
Weekly trends reinforce the same seasonality seen in the daily data, but with clearer segmentation of active and inactive periods. These weekly peaks are crucial for high-level resource planning since they show the

exact weeks when operations must be fully staffed and stocked.



Weekly order totals were calculated by aggregating daily orders by week. **Early Semester (Weeks 1–2):** Orders were small, showing limited campus activity at the start of the semester. **Mid Semester Peak (Weeks 3–10 and 12–18):** Weekly totals consistently exceeded 16,500 items, with mean daily orders between 2,400–2,800. These are the weeks when the campus was fully occupied, and there was a strong demand for dining. **Low Activity / Break Weeks (Weeks 11, 19–22):** Weeks with significantly lower totals representing the times when the spring break, holidays, or the end of the semester occurred, and thus fewer students were on campus.

The monthly time series shows a classic academic calendar pattern: moderate early-semester usage, peak mid-semester demand, and steep declines during academic breaks and after final exams. This long-term trend underscores the importance of aligning staffing and inventory with the academic cycle.



6. Regression Analysis

The regression analysis was conducted only for items sold at Panera Bread, which had the highest sales during the Spring 2025 semester.

6.1 Nutrition Regression Analysis

Quantity against Nutrition: A linear regression was estimated with average daily quantity sold per item as the dependent variable and item-level nutrition measures (total fat, carbohydrates, protein, and sodium, all per serving) as predictors (code snippet in Figure 4). The model's output (Figure 5) shows an adjusted R^2 is 0.018, indicating that these four nutritional variables together explain only about 2% of the variation in sales across Panera items.

Among the coefficients, only sodium is statistically significant at the 5% level. The estimated coefficient of approximately 0.0046 means that, holding fat, carbohydrates, and protein constant, an additional 1,000 mg of sodium is associated with an increase of roughly 4–5 units in average daily quantity sold. This association appears to be driven by a few very popular, high-sodium items. For example, several Mac & Cheese–based entrées rank among the top sellers (Figure 6) and contain up to about 3,000 mg of sodium per serving, whereas many lower-sodium items sell fewer units. However, given the low overall explanatory power of the model and the potential multicollinearity among nutrients, the sodium effect could not be interpreted as a causal impact on demand.

Quantity against Item Category: To complement the nutrition analysis, a second regression (Figure 7) was estimated with average daily quantity as the dependent variable and item category (e.g., soups, salads, bakery) as the sole predictor which shows that menu structure is an important driver of sales. The model achieves an adjusted R^2 of 0.34, indicating that categories alone explain roughly one-third of the variation in per-item daily sales. The “You Pick 2” category stands out with a coefficient of 28.5 ($p =$

0.0003), meaning that “You Pick 2” items sell on the order of 28 more units per day than items in the typical category. Combined with the descriptive counts by category (Figure 8), this implies that combination of meals—rather than individual soups, salads, or sandwiches—are the primary engine of Panera volume.

Popular high-sodium offerings, such as Mac & Cheese bowls that frequently appear in these bundles, likely explain the positive sodium association observed in the nutrition model, reinforcing the interpretation that customers are mainly responding to attractive bundled meal formats rather than to specific nutrient information.

6.2 Weather Regression Analysis

The items in the product mix were aggregated by date and summarized by total quantities sold per day at each venue, before merging with the weather dataset to conduct the analysis (code snippet in Figure 10). A PCA analysis (Figure 9) was conducted to identify that one of the temperature variables and one of the precipitation related variables significantly contributed to the two dimensions.

A linear regression of daily Panera sales (Figure 11) on temperature, humidity, precipitation, and snowfall is significant but explains only about 8% of variation in sales (adjusted $R^2 \approx 0.08$). Temperature shows no effect ($p = 0.96$). Humidity has a strong negative association: each 1-point increase in humidity corresponds to about 8 fewer items sold per day ($p = 0.0017$). Precipitation has a positive, marginally significant coefficient ($\approx +282$ items per inch, $p \approx 0.066$), suggesting rainy days may increase sales, though the estimate is uncertain. Snowfall has a negative, marginally significant effect (≈ -110 items per inch, $p \approx 0.063$), consistent with snow reducing campus activity and traffic. Overall, weather influence on demand is marginal, with more than 90% of daily sales variation driven by non-weather factors.

6.3 Day-of-the-Week Regression Analysis

Daily Panera sales were regressed on day-of-week indicators to assess recurring weekly patterns (Figure 12). This model shows that day of week has a modest relationship with Panera's daily sales, with Saturdays notably weaker (~235 items fewer) than weekdays, but overall day-of-week patterns explain only a small share (4%) of variation.

Amongst the Top 3 highest-volume categories (Figure 16), day-of-week regressions (Figure 13, 14, 15) reveal strong and consistent weekly patterns. For "You Pick 2" and "Breakfast," day of week alone explains roughly half of the variation in daily sales (adjusted $R^2 \approx 0.50$), with similar but slightly weaker effects for "Soups or Mac & Cheese Bowl" (adjusted $R^2 \approx 0.20$). In all three categories, weekday volumes (Monday–Thursday, and often Friday) are broadly comparable to Wednesday, while weekends show pronounced declines. Relative to Wednesday, "You Pick 2" sales are lower by about 52 units on Fridays, 150 units on Saturdays, and 63 units on Sundays; breakfast items are lower by about 42 units on Saturdays and 68 units on Sundays; and soup/Mac & Cheese bowls are lower by about 29 units on Saturdays and 20 units on Sundays. These patterns confirm that Panera's highest-volume categories are heavily concentrated on weekday traffic, particularly mid-week lunch and breakfast, with much lower engagement on weekends.

7. Forecasting

Forecasting models were utilized to identify trends and patterns of both the top three items, and of the Panera Bread venue. Outlier dates were omitted, including spring break, the time period before the University opened in Jan, and the time of post-finals. Since we can't detect long-term trends due to our time-limited data frame (ie. $\beta = \text{false}$), but clearly has seasonality, a double-exponential smoothing forecast model is used: a choice further supported by its consistent outperformance of the other models by having the lowest RMSE score.

7.1 Forecast of Top Three Items Sold

Qdoba Bowls are ordered ~163 times a day on average, with $\alpha = 0.02$ and $\gamma = .10$ (Figure 18).

This implies that the model adjusts very slowly with new data, is quite stable, and doesn't predict significant changes in seasonality/overall change. The RMSE is ~22.7, implying our model was off, on average, by 22.7 bowls, based on our training data. Tuesdays are the worst day for Qdoba bowl sales with a ~-47.51 coefficient, and Thursdays are the best day with a 25.17 coefficient.

Panera's half sandwich & soup cup is ordered ~121 times a day on average, with $\alpha = 0.00$, and $\gamma = .35$ (Figure 19). This implies that the model will most likely be unimpacted by a new data point, except when catering to changes in seasonality shifts. The RMSE is ~28.75, implying our model was off, on average, by 28.75 orders per day, based on our training data. Sundays are the worst for the quantity of Panera's deal sales with a ~-13.75 coefficient, and Fridays are the best day with a 22.21 coefficient.

True Burgers' three chicken tenders are ordered ~176 times a day on average, with an α of .16 and a γ of .56 (Figure 20). Although this may initially seem quite significant, the RMSE is ~60.98, and the RMSE being greater than a third of the mean orders implies that this forecast isn't too accurate and should be approached with caution.

7.2 Panera Bread Forecast

The double-forecasting model output applied to all of Panera Bread's aggregate quantity output indicated that Panera sells around ~730 items a day (Figure 21). Tuesdays are their worst day for quantity of items sold, with a -211.04 coefficient, while Saturday was their best date, with a 181.55 coefficient. Its α is ~.03 and γ is ~.57, implying the base model is quite hesitant to fundamentally shift its forecasts on new data, but recognizes, and would quite substantially shift with γ being so high, implying Panera has noticeable, changing weekly trends the model is picking up. With an RMSE of ~129.51, this forecasting method indicates a higher level of confidence.

APPENDIX

Figure 1: Product Mix Sample Data

Date	Venue	Category	Item	Quantity
2/21/2025	ACE Sushi	Poke Bowl	Spicy Tuna Bowl	2
2/21/2025	ACE Sushi	Poke Bowl	Vegan Bowl	1
2/21/2025	ACE Sushi	Poke-ME	Poke Bowl	30
2/21/2025	ACE Sushi	Sushi-ME	Spicy California Roll	4
2/21/2025	ACE Sushi	Sushi-ME	Spicy Tuna Roll	11
2/21/2025	ACE Sushi	Sushi-ME	The Top Trio Combo	2
2/21/2025	ACE Sushi	Sushi-ME	Vegetarian Roll	5
2/21/2025	ACE Sushi	Tea-ME	Guava	1
2/21/2025	ACE Sushi	Tea-ME	Peach Mango	3
2/21/2025	ACE Sushi	Tea-ME	Raspberry Lemonade	2
2/21/2025	ACE Sushi	Tea-ME	Taro Milk Tea	2
2/21/2025	ACE Sushi	Tea-ME	Thai Tea	3
2/21/2025	ACE Sushi	Sushi Roll	Tempura Crunch Roll	1
2/21/2025	ACE Sushi	Sushi Roll	Tempura Shrimp Roll	1
2/21/2025	ACE Sushi	Sushi Roll	Avocado Cucumber Crunch Roll	1
2/21/2025	ACE Sushi	Sushi Roll	Vegetarian Crunch Roll	1
2/21/2025	Freshens	Smoothies	All Fruit Blends	9
2/21/2025	Freshens	Salads	Chicken Caesar	3
2/21/2025	Freshens	Rice Bowls	Buffalo RB	2
2/21/2025	Freshens	LTO	Chicken Quesadilla	2
2/21/2025	Freshens	Artisan Melts	Buffalo Chicken	13
2/21/2025	Freshens	Artisan Melts	Chicken Bacon Ranch	21
2/21/2025	Freshens	Artisan Melts	Chicken Caprese	8
2/21/2025	Freshens	Crunchadillas	Chicken Avocado	19
2/21/2025	Freshens	Crunchadillas	Chicken Bacon Ranch C	15

Figure 2: Nutrition Sample Data

Item	Calories	Total_fat (grams)	Carbs (grams)	Protein (grams)	Sodium (mgrams)
Agave Lemonade	210	0	52	0	15
Americano	5	0	0	1	15
Apple	80	0	22	0	0
Artisan Ciabatta Loaf	150	1	31	6	300
Asiago Bagel	320	5	55	14	540

Asiago Cheese Bagel	320	5	55	14	540
Bacon Double Take	430	25	30	21	890
Bacon Mac & Cheese	860	45	87	34	2440
Banana	110	0	29	1	0
Bistro French Onion Soup	370	13	44	15	1620
Black Pepper Focaccia	160	3	30	5	440
Blueberry Bagel	290	1	61	10	390
Blueberry Lavender Lemonade	220	0	55	0	20

Figure 3: Weather Sample Data

name	datetime	tempmax	tempmin	temp	feelslike	feelslike	feelslike	dew	humidity	precip	precippro	precipcov	precipcov	precipcov	snow	snowdept	windgust	windspeed	winddir	sealevel	cloudcover	visibility	solarradi	solarener	uvindex
Washington, DC	1/1/2025	51	40.8	47.2	51	36.2	42.9	31.8	57.2	0.01	100	4.17	rain	0	0	0	53	29.6	278.8	1008.2	75.2	9.8	88.3	7.7	5
Washington, DC	1/2/2025	44.7	37.8	39.8	37.9	29.4	33	16.8	39	0	0	0		0	0	0	38.5	18.3	293.4	1020.1	67.7	9.9	74.2	6.5	4
Washington, DC	1/3/2025	44.8	30.7	36	38	18.9	29.3	21.1	54.9	0.037	100	12.5	rain	0	0	0	37.2	18.2	270.1	1016	75.5	9.5	45	3.9	3
Washington, DC	1/4/2025	33.8	26.8	30.3	22.2	16.5	19.2	9.5	41.6	0	0	0		0	0	0	44.3	24	294.4	1019.1	46.1	9.9	63.5	5.5	5
Washington, DC	1/5/2025	37	23.9	31	35.6	15.9	24.8	6.3	36	0	0	0		0	0	0	23.5	13.9	294.9	1021.9	57.4	9.9	87.6	7.7	5
Washington, DC	1/6/2025	33.9	28	29.2	28.9	17.5	21	23.6	80.1	0.844	100	87.5	rain, snow	6.2	11.5	20.8	14.8	18.6	1010.9	99.6	3	13.8	1.2	1	
Washington, DC	1/7/2025	35	21.9	28.9	29	9.2	17.3	15.5	57.4	0	0	0		0	0	9.5	36.6	25.8	321.3	1017.6	47.2	9.9	106.3	9.3	5
Washington, DC	1/8/2025	33.9	25.9	29.1	34.7	13.3	18.6	11.7	48.1	0	0	0		0	0	7.5	34	19.7	313.5	1019.6	82.3	9.9	94	8.3	7
Washington, DC	1/9/2025	33.9	22.9	28.2	23.5	10.9	16.8	6.9	40.2	0	0	0		0	0	8.8	45	28.6	315	1017.7	12.5	9.9	121.1	10.5	5
Washington, DC	1/10/2025	34.9	24.9	30	33.8	11.8	22.3	8.7	40.5	0	0	0		0	0	8	27.6	17	313.1	1017.1	65	9.9	88.3	7.8	4
Washington, DC	1/11/2025	38.7	28.9	32.9	29.8	20.4	25.3	20.7	62.2	0.059	100	16.67	rain, snow	0.3	6.9	32.1	18.4	308.5	1005.9	79.3	8.5	89.6	7.7	5	
Washington, DC	1/12/2025	42.1	27.6	34.6	42.1	21.6	30.2	20.4	56.8	0	0	0		0	0	3.4	15.7	9	239.6	1017.9	32.2	9.8	119.4	10.3	5
Washington, DC	1/13/2025	49.8	28.8	37.7	48.9	22.9	33.7	22.1	54.6	0	0	0		0	0	2.4	18.6	9.1	227.4	1016.9	51.3	9.9	73.9	6.4	5
Washington, DC	1/14/2025	37.8	28.9	33	35	15.6	24.3	9.8	37.9	0	0	0		0	0	1	34.2	25.1	317.1	1020.6	30.4	9.9	122	10.5	5
Washington, DC	1/15/2025	33	22.9	27.5	24.2	11.6	17	8.1	43.4	0	0	0		0	0	0.1	33.6	20.7	304.9	1028.5	10.3	9.9	123.9	10.8	5
Washington, DC	1/16/2025	37	20.5	28.7	33.1	13.1	22.2	16.1	59.2	0.031	100	4.17	rain, snow	0	0	0	18.6	13.6	211.4	1016.5	71	9.6	55.9	4.7	3
Washington, DC	1/17/2025	45.5	30.6	37.2	43.2	24.4	32.4	23.8	59.2	0	0	0		0	0	20.4	12.7	225.7	1019.1	53.1	9.9	75.5	6.4	5	
Washington, DC	1/18/2025	44.9	34	38.2	40.8	27.2	33.6	30.1	72.7	0.038	100	4.17	rain	0	0	0	20.1	11.3	192.8	1012.5	69.2	9.9	29	2.4	1
Washington, DC	1/19/2025	40.9	28.9	35.7	37.8	17.3	29.5	30.1	80	0.219	100	20.83	rain, snow	0.4	0	0	35.7	22.4	350.2	1009.4	95.2	7.6	11.9	1	1

Figure 4: Code Snippet for Nutrition Regression Analysis

```
#Read the product mix data
productmix <- read.csv("productmixv2.csv")
productmix <- as.data.frame(productmix)
productmix$Date <- as.Date(productmix$Date, format = "%m/%d/%Y")

#Read the nutrition data
nutrition <- read.csv("nutrition_facts.csv")
nutrition <- as.data.frame(nutrition)
nutrition$Item <- trimws(nutrition$Item)

#Filter, aggregate and summarize the product mix data
productmix <- productmix[productmix$Venue == "Panera Bread",]
item_wide <- productmix %>%
  group_by(Item) %>%
  summarise(
    quantity = mean(Quantity))

#Merge nutrition data with the product mix data
merged_df <- merge(item_wide, nutrition, by.x = "Item", by.y = "Item")

#Run linear regression analysis
lm_result <- lm(quantity ~ Total_fat_grams + Carbs_grams + Protein_grams + Sodium_mgrams, data = merged_df)
```

Figure 5: Nutrition Regression Analysis Output

```
Call:
lm(formula = quantity ~ Total_fat_grams + Carbs_grams + Protein_grams + Sodium_mgrams, data = merged_df)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-9.479 -2.845 -1.738 -0.684  80.769

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.012682   1.397139   1.441   0.1517
Total_fat_grams 0.046673   0.088686   0.526   0.5994
Carbs_grams    -0.006360   0.029158  -0.218   0.8276
Protein_grams  -0.163032   0.118467  -1.376   0.1707
Sodium_mgrams  0.004575   0.002144   2.134  0.0344 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.454 on 160 degrees of freedom
Multiple R-squared:  0.04215, Adjusted R-squared:  0.0182
F-statistic: 1.76 on 4 and 160 DF, p-value: 0.1395

```

Figure 6: Top Items sold at Panera Bread (by Avg Quantities per Day)

Item	Calories	Fat(g)	Carbs(g)	Protein(g)	Sodium(mg)	Avg Qty
Half Sandwich and Cup of Soup	400	15	35	11	700	84.7
Mac & Cheese and Half Sandwich	690	35	67	23	1600	67.5
Half Sandwich and Half Salad	420	18	36	13	710	40.3
Large Mac & Cheese	1070	55	111	43	3040	28.4
Bacon Double Take	430	25	30	21	890	22.5
Garden Avo & Egg White Sandwich	290	11	31	14	460	19.5

Figure 7: Regression of Quantities against Item Categories at Panera

```

Call:
lm(formula = quantity ~ Category, data = merged_df)

Residuals:
    Min       1Q   Median       3Q      Max
-28.131 -0.613 -0.121  0.232  55.579

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.58108    7.30399   0.080 0.936695
CategoryBagels  1.93581    7.88921   0.245 0.806498
CategoryBakery  2.23311    8.94553   0.250 0.803207
CategoryBeverages -0.57432   10.32940  -0.056 0.955733
CategoryBreakfast  7.47557    7.57971   0.986 0.325572
CategoryBreakfast Sandwiches  0.03089    7.80830   0.004 0.996849
CategoryCold Coffees  5.00676   10.32940   0.485 0.628580
CategoryCold Drinks  0.97095    7.66049   0.127 0.899308
CategoryCookies & Brownies  0.44459    7.66049   0.058 0.953795
CategoryDrinks  -0.41892   10.32940  -0.041 0.967703
CategoryFrozen Drinks & Smoothies  1.46081    8.00112   0.183 0.855374
CategoryHalf Salads -0.33277    7.60224  -0.044 0.965143
CategoryHalf Sandwiches -0.02660    7.52878  -0.004 0.997185
CategoryHot Beverages  3.14865   10.32940   0.305 0.760917
CategoryHot Coffees -0.40541   10.32940  -0.039 0.968744
CategoryHot Drinks & Espresso  0.21419    7.66049   0.028 0.977731
CategoryIced Drinks & Espresso  0.85923    7.88921   0.109 0.913416
CategoryIced Teas  6.46622   10.32940   0.626 0.532253
CategoryMac & Cheese  1.91892    8.43392   0.228 0.820322
CategoryMuffins & Muffies  1.76351    8.94553   0.197 0.843982

```

CategoryOatmeal, Yogurt & Fruit	-0.32432	10.32940	-0.031	0.974993
CategoryPastries	1.23198	8.43392	0.146	0.884056
CategoryScones	0.87838	10.32940	0.085	0.932344
CategorySides	2.12838	10.32940	0.206	0.837027
CategorySides & Spreads	-0.20158	7.88921	-0.026	0.979649
CategorySoups	0.50976	7.69908	0.066	0.947297
CategorySoups or Mac & Cheese Bowl	9.44764	8.16611	1.157	0.249115
CategoryTake Home Loaves and Packs	-0.41667	7.88921	-0.053	0.957949
CategoryValue Duet	0.78829	8.43392	0.093	0.925656
CategoryWhole Salads	-0.28435	7.60224	-0.037	0.970213
CategoryWhole Sandwiches	0.46300	7.47587	0.062	0.950698
CategoryYou Pick 2	28.50929	7.74705	3.680	0.000323 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error: 7.304 on 152 degrees of freedom				
Multiple R-squared: 0.4524, Adjusted R-squared: 0.3407				
F-statistic: 4.051 on 31 and 152 DF, p-value: 3.866e-09				

Figure 8: Avg Quantity per Day by Item Category at Panera Bread

Category	Avg Quantity
1 "Bagels"	15.10
2 "Breakfast "	146.00
3 "Breakfast Sandwiches"	9.36
4 "Cold Drinks"	15.50
5 "Cookies & Brownies"	10.30
6 "Frozen Drinks & Smoothies"	10.20
7 "Half Salads"	2.98
8 "Half Sandwiches"	8.87
9 "Hot Drinks & Espresso"	8.41
10 "Iced Drinks & Espresso "	8.64
11 "Mac & Cheese"	7.50
12 "Muffins & Muffies"	4.69
13 "Oatmeal, Yogurt & Fruit"	0.26
14 "Pastries "	6.29
15 "Scones "	1.46
16 "Sides & Spreads"	2.28
17 "Soups"	9.82
18 "Soups or Mac & Cheese Bowl"	40.10
19 "Take Home Loaves and Packs"	0.99
20 "Value Duet"	4.11
21 "Whole Salads"	3.56
22 "Whole Sandwiches"	22.00
23 "You Pick 2"	233.00

Figure 9: PCA Analysis

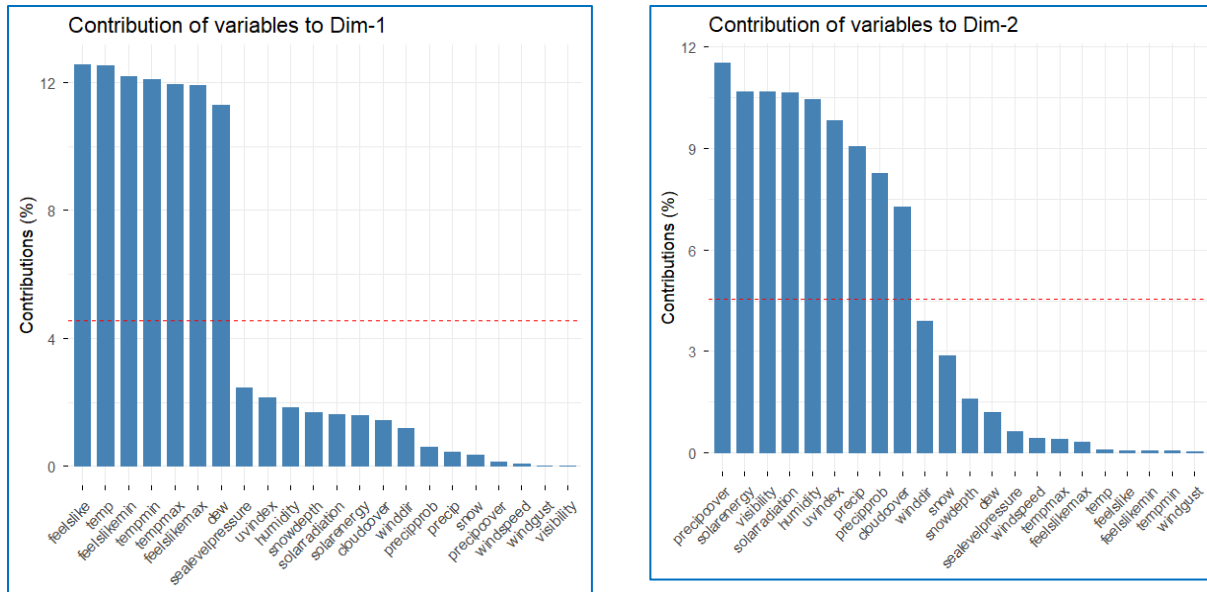


Figure 10: Code Snippet of Weather Regression Analysis

```
#Aggregate by date as rows with Venue as columns
venue_wide <- productmix %>% group_by(Venue, Date) %>%
  summarise(total_quantity = sum(Quantity), .groups = 'drop') %>%
  pivot_wider(names_from = Venue,
    values_from = total_quantity,
    values_fill = 0)

#Each column is a venue's daily quantity
head(venue_wide)
summary(venue_wide$Panera Bread)

#Merge the productmix and weather dataframes
merged_df <- merge(venue_wide, weather, by.x = "Date", by.y = "datetime")
colnames(merged_df)[colnames(merged_df) == "Panera Bread"] <- "Panera"

#Regress quantity against weather factors
lm_result <- lm(Panera ~ temp + humidity + precip + snow, data = merged_df)
summary(lm_result)
```

Figure 11: Weather Regression Analysis Output

```
Call:
lm(formula = Panera ~ temp + humidity + precip + snow, data = merged_df)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-734.7 -181.2  110.7  249.9  548.3

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1032.9414   152.8889   6.756 3.95e-10 ***
temp          0.1199    2.2343    0.054 0.95727
humidity     -8.2018    2.5585   -3.206 0.00168 **
precip       281.5970   151.6347    1.857 0.06550 .
snow        -109.7194    58.5312   -1.875 0.06303 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 356.9 on 134 degrees of freedom
Multiple R-squared:  0.1053,    Adjusted R-squared:  0.07856
F-statistic: 3.941 on 4 and 134 DF,  p-value: 0.004671

```

Although precipitation and humidity are both weather variables, in this dataset they are only weakly correlated (correlation ≈ 0.23), and their variance inflation factors are low, indicating that they capture largely distinct dimensions of daily weather conditions.

Figure 12: Day-of-Week Regression Analysis Output

```

Call:
lm(formula = Panera ~ weekday_Fri + weekday_Mon + weekday_Sat +
    weekday_Sun + weekday_Thu + weekday_Tue + weekday_Wed, data = merged_df)

Residuals:
    Min       1Q   Median       3Q      Max
-722.5 -198.1  160.9  242.3  360.1

Coefficients: (1 not defined because of singularities)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   648.30     81.47   7.958 7.02e-13 ***
weekday_Fri   -95.82    113.83  -0.842  0.4014
weekday_Mon    61.40    115.21  0.533  0.5950
weekday_Sat  -235.25    116.72  -2.016  0.0459 *
weekday_Sun  -136.69    118.37  -1.155  0.2503
weekday_Thu    27.60    113.83  0.243  0.8088
weekday_Tue    74.15    115.21  0.644  0.5210
weekday_Wed      NA         NA     NA     NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 364.3 on 132 degrees of freedom
Multiple R-squared:  0.08138, Adjusted R-squared:  0.03962
F-statistic: 1.949 on 6 and 132 DF,  p-value: 0.07753

```

Figure 13: Day-of-Week Regression Analysis Output for Category 1

```
> summary(models[["You Pick 2"]])
```

```
Call:
lm(formula = formula, data = categ_wide)

Residuals:
    Min       1Q   Median       3Q      Max
-299.33  -11.50   12.12   28.41   89.67

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  352.333     14.758   23.875 < 2e-16 ***
weekday_Fri  -52.000     20.870   -2.492  0.01438 *
weekday_Mon    9.542     20.542    0.465  0.64331
weekday_Sat -149.833     21.240   -7.054 2.39e-10 ***
weekday_Sun  -63.200     20.870   -3.028  0.00314 **
weekday_Thu   34.133     20.870    1.636  0.10512
weekday_Tue   16.042     20.542    0.781  0.43670
weekday_Wed      NA           NA      NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 57.16 on 99 degrees of freedom
Multiple R-squared:  0.5256, Adjusted R-squared:  0.4968
F-statistic: 18.28 on 6 and 99 DF, p-value: 3.484e-14
```

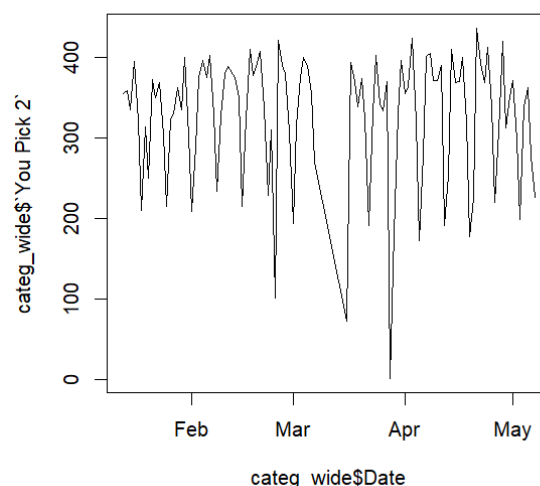


Figure 14: Day-of-Week Regression Analysis Output for Category 2

```
> summary(models[["Breakfast "]])

Call:
lm(formula = formula, data = categ_wide)

Residuals:
    Min       1Q   Median       3Q      Max
-92.200 -14.150   4.333  17.228  55.786

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  212.667     7.288   29.179 < 2e-16 ***
weekday_Fri    7.200     10.307    0.699  0.486478
weekday_Mon    9.583     10.145    0.945  0.347140
```

```

weekday_Sat -42.452    10.490  -4.047 0.000103 ***
weekday_Sun -68.467    10.307  -6.643 1.68e-09 ***
weekday_Thu  13.733    10.307   1.332 0.185788
weekday_Tue   9.896    10.145   0.975 0.331715
weekday_Wed    NA         NA     NA     NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.23 on 99 degrees of freedom
Multiple R-squared:  0.5406, Adjusted R-squared:  0.5128
F-statistic: 19.42 on 6 and 99 DF,  p-value: 7.438e-15

```

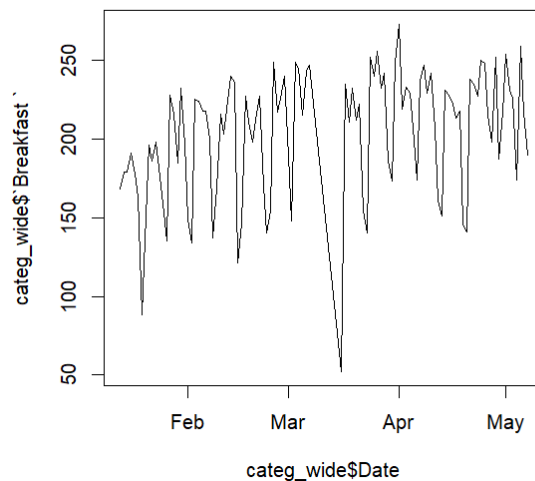


Figure 15: Day-of-Week Regression Analysis Output for Category 3

```

> summary(models[["Soups or Mac & Cheese Bowl"]])

Call:
lm(formula = formula, data = categ_wide)

Residuals:
    Min       1Q   Median       3Q      Max
-63.875 -14.475   1.314  15.813  97.125

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   60.933     6.799   8.962 2.01e-14 ***
weekday_Fri  -14.133     9.616  -1.470  0.14478
weekday_Mon    8.942     9.464   0.945  0.34707
weekday_Sat  -29.362     9.786  -3.000  0.00341 **
weekday_Sun  -20.267     9.616  -2.108  0.03759 *
weekday_Thu    5.467     9.616   0.569  0.57098
weekday_Tue   11.254     9.464   1.189  0.23724
weekday_Wed    NA         NA     NA     NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.33 on 99 degrees of freedom
Multiple R-squared:  0.2461, Adjusted R-squared:  0.2004
F-statistic: 5.386 on 6 and 99 DF,  p-value: 7.513e-05

```

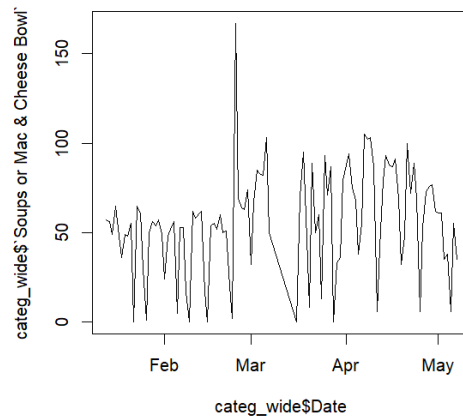


Figure 16: TopN Categories by Avg Quantity Sales at Panera

```
> print(categ_wide)
Category               quantity
1 "You Pick 2"          233.00
2 "Breakfast "          146.00
3 "Soups or Mac & Cheese Bowl" 40.10
4 "Whole Sandwiches"    22.00
5 "Cold Drinks"         15.50
6 "Bagels"              15.10
7 "Cookies & Brownies"   10.30
8 "Frozen Drinks & Smoothies" 10.20
9 "Soups"                9.82
10 "Breakfast Sandwiches" 9.36
```

Figure 17: A three-way time-series analysis comparing the quantity of items sold (D/W/M)

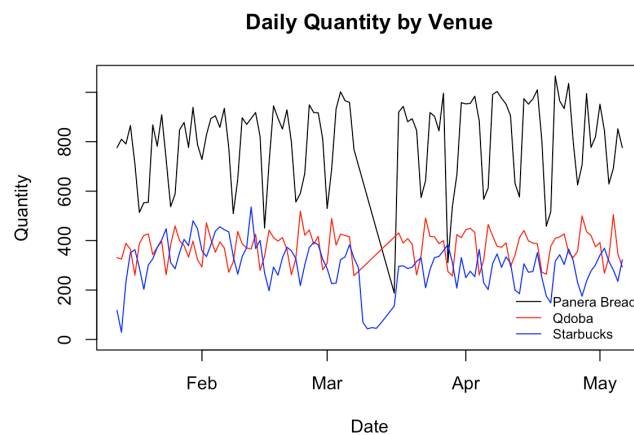


Figure 18: A DES forecasting model of Qdoba Bowls

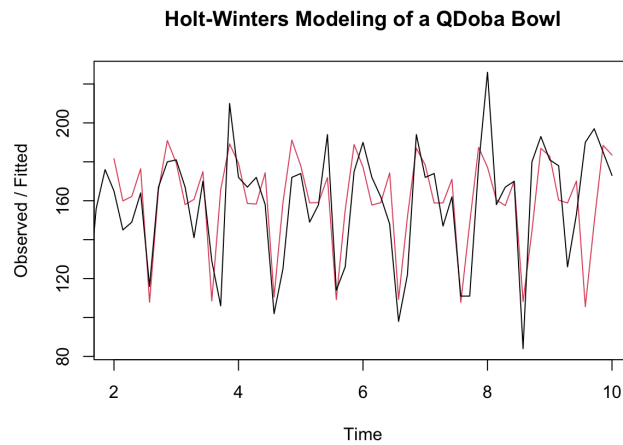


Figure 19: A DES forecasting model of Panera's Half Sandwich & Soup Cup

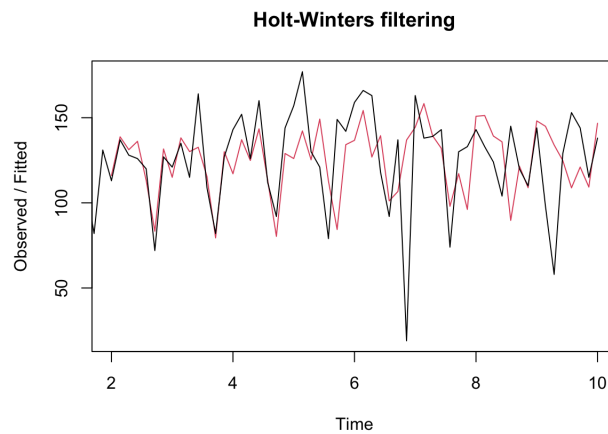


Figure 20: A DES forecasting model of Panera's Half Sandwich & Soup Cup

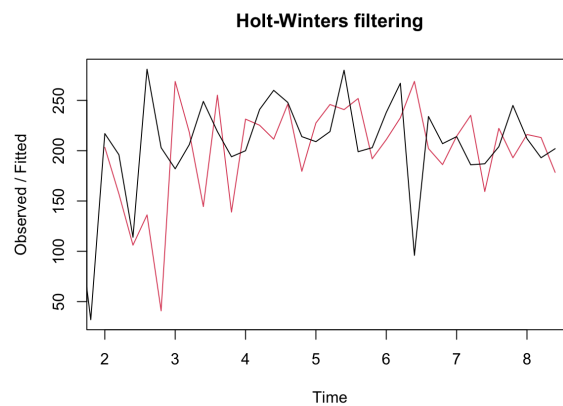


Figure 21: A DES forecasting model of Panera's aggregate Quantity sales

